

SCULPT: Training Edge Vision Models for Post-Training Quantization Readiness

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Abstract

Edge vision models are difficult to deploy on resource-constrained hardware, making low-bit post-training quantization (PTQ) attractive. In practice, standard FP32 training often produces heavy-tailed activation distributions whose outliers destabilize activation quantization: preserving the full range wastes quantization bins on rare extremes, while aggressive clipping causes information loss. Existing solutions typically rely on quantization-aware training (QAT), which adds training complexity and bit-width coupling, or advanced PTQ procedures that repair the model after training.

We present SCULPT (Statistical Clipping and Uniform Loss for Post-Training), a training-time method that improves PTQ readiness during ordinary FP32 fine-tuning. SCULPT combines a topology-aware activation regularizer that suppresses quantization-hostile skewness and kurtosis with a stable percentile-based clipping mechanism that learns deployment-ready activation bounds. Unlike QAT, SCULPT does not simulate quantization during optimization; unlike post hoc outlier-repair PTQ methods, it does not require runtime activation transformations. The learned clipping bounds can be exported directly into a standard PTQ workflow for low-bit deployment, including INT8 and lower-bit settings such as W4A8.

Keywords: Post-training quantization, Activation outliers, Edge vision, Model compression

1 Introduction

Edge vision and perception models remain difficult to deploy on resource-constrained hardware due to tight latency, memory, and power budgets. Low-bit quantization is attractive because it enables efficient integer inference while preserving existing model architectures and deployment pipelines. In practice, however, post-training quantization (PTQ) often degrades sharply at low precision when activations exhibit heavy-tailed or highly asymmetric distributions. If the quantizer preserves rare outliers, the effective resolution for the dense bulk of the distribution becomes too coarse; if it clips aggressively, those rare extremes are truncated [Banner et al., 2019].

Figure 1 illustrates this failure mode on a representative hard layer. Most activations remain concentrated in a narrow operating region, but a small number of extreme values drive the maximum far beyond the normal scale of the layer. This mismatch makes min/max calibration brittle and is a major source of PTQ error.

Existing methods typically address this problem either during quantized training or after the full-precision model has already been trained. Quantization-aware training (QAT) and related approaches learn clipping thresholds or quantizer parameters during optimization, but they add training complexity and are often tied to a specific quantization setting [Choi et al., 2018, Jain et al., 2020, Esser et al., 2020]. On the other hand, advanced PTQ methods improve low-bit accuracy through post hoc calibration, adaptive rounding, reconstruction, or activation smoothing after training [Nagel et al., 2020, Li et al., 2021, Xiao et al., 2023]. These methods can be effective, but they also add workflow complexity or transformations that are unattractive in deployment-constrained settings.